Richie B. Thomas

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Education

Bachelor of Engineering - Information Technology **2022 – 2026**
Fr. C. Rodrigues Institute of Technology CGPA: 7.42/10.0

Professional Experience

Research Intern **July 2025 – Present**
Indian Institute of Science Education and Research (IISER), Mohali, India
- Spearheaded hardware-software co-design of a Sparse Neural Network Accelerator on PYNQ-Z2 FPGA for MNIST digit recognition
- Optimized neural architecture achieving 96% accuracy, enabling real-time edge AI deployment

Tech Intern **July 2025 – July 2025**
Sports Mentoring Infusion, Mumbai
- Architected cross-platform React Native application with Node.js backend, reducing load times by 40% through optimized API design
- Launched gamified habit tracking system that boosted user engagement by 65% through streak rewards and progress visualization
- Engineered analytics dashboard enabling data-driven coaching decisions for 15+ sports mentors

Android Developer Intern **June 2025 – July 2025**
GetFly Technologies, Sion
- Delivered campus ERP solution using Flutter, digitizing operations for 1,000+ students and faculty across 8 modules at Vasantdada Patil Pratishthan's College of Engineering & Visual Arts (VPPCOEVA)
- Established unified project architecture by consolidating individual module codebases from multiple developers into a cohesive, scalable application
- Integrated 8+ REST APIs and optimized Dio-based data fetching, reducing sync latency by 40%

Full Stack Web Developer Intern **March 2025 – June 2025**
Bayside Sports India Pvt. Ltd., Bandra
- Built SSR sales portal with React.js/Laravel, reducing initial page load time by 50% and enhancing crawlability
- Streamlined booking workflow cutting customer onboarding time by 35% through UI/UX optimization
- Developed admin dashboard that cut lead qualification time by 40%, allowing sales reps to tailor outreach based on user booking data

PHP Developer **March 2024 – April 2025**
Fr. C. Rodrigues Institute of Technology, Vashi
- Owned campus management system serving 1,000+ users with RBAC
- Automated data exports saving 10+ administrative hours weekly through PDF/Excel modules
- Drove 20% UX improvement via performance tuning and feature enhancements

Key Projects

Investment Platform - MERN Stack GitHub
Built in 24 hours during a hackathon by a 4-member team, this end-to-end trading platform features portfolio management and automated trades. Integrated real-time data streams via WebSockets, increasing data freshness by 5x. Tech stack: MongoDB, Express.js, React.js, Node.js, Chart.js.

Learning Management System - Flutter & Firebase GitHub
Education platform enabling content delivery and grade tracking. Incorporated Firebase Auth reducing auth failures by 70%

Technical Skills

Languages:	TypeScript, JavaScript (ES6+), Python, Java, Dart, SQL
Frontend:	React.js, Flutter, React Native, Material UI, Chart.js
Backend:	Node.js, Express, REST/GraphQL, WebSockets, Server-Side Rendering
Databases:	PostgreSQL, MongoDB, Firebase, ORM (Mongoose, Eloquent)
DevOps:	AWS, Docker, CI/CD, Git, Firebase, Postman
Methodologies:	Agile, TDD, Microservices, Hardware-Software Co-Design

Leadership & Activities

Head of Documentation **July 2024 – March 2025**
Computer Society of India (CSI)
- Managed documentation for 10+ technical events with 500+ participants, creating standardized templates and automated certification system in Python
- Improved operational efficiency by 40% through implementation of document management workflow